HEALTH MONITORING NODE PROGRAM-2

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➔ **Source Code:**

```
/*
  Formal Design of Embedded Controller Using FSM
  Traffic Light Controller
  Arduino UNO + Proteus
*/
enum TrafficState {
  STATE_NS_GREEN,
  STATE_NS_YELLOW,
  STATE_EW_GREEN,
  STATE_EW_YELLOW
};
TrafficState currentState = STATE_NS_GREEN;
// North-South traffic signal pins
const int NS_RED_PIN    = 2;
const int NS_YELLOW_PIN = 3;
const int NS_GREEN_PIN  = 4;
// East-West traffic signal pins
```

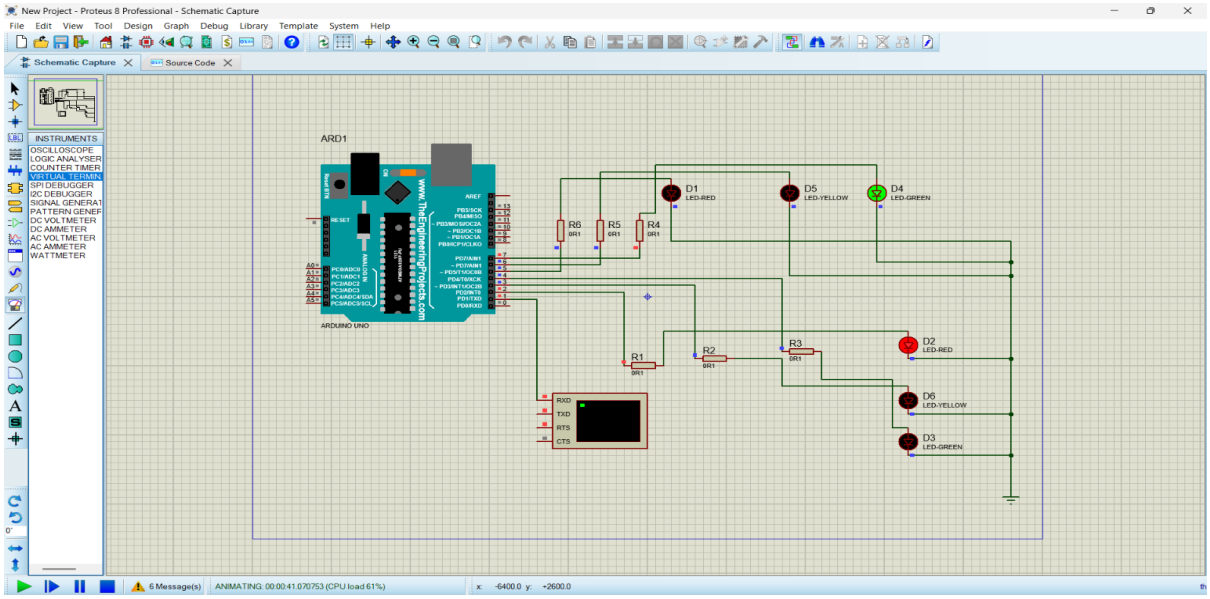
```
const int EW_RED_PIN    = 5;
const int EW_YELLOW_PIN = 6;
const int EW_GREEN_PIN  = 7;
void allOff() {
  digitalWrite(NS_RED_PIN, LOW);
  digitalWrite(NS_YELLOW_PIN, LOW);
  digitalWrite(NS_GREEN_PIN, LOW);
  digitalWrite(EW_RED_PIN, LOW);
  digitalWrite(EW_YELLOW_PIN, LOW);
  digitalWrite(EW_GREEN_PIN, LOW);
}
void enterState(TrafficState state) {
  allOff();
  switch (state) {
    case STATE_NS_GREEN:
      digitalWrite(NS_GREEN_PIN, HIGH);
      digitalWrite(EW_RED_PIN, HIGH);
      Serial.println("S0: NS GREEN - EW RED");
      break;
    case STATE_NS_YELLOW:
      digitalWrite(NS_YELLOW_PIN, HIGH);
      digitalWrite(EW_RED_PIN, HIGH);
      Serial.println("S1: NS YELLOW - EW RED");
```

```
      break;
    case STATE_EW_GREEN:
      digitalWrite(NS_RED_PIN, HIGH);
      digitalWrite(EW_GREEN_PIN, HIGH);
      Serial.println("S2: NS RED - EW GREEN");
      break;
    case STATE_EW_YELLOW:
      digitalWrite(NS_RED_PIN, HIGH);
      digitalWrite(EW_YELLOW_PIN, HIGH);
      Serial.println("S3: NS RED - EW YELLOW");
      break;
  }
}
void setup() {
  pinMode(NS_RED_PIN, OUTPUT);
  pinMode(NS_YELLOW_PIN, OUTPUT);
  pinMode(NS_GREEN_PIN, OUTPUT);
  pinMode(EW_RED_PIN, OUTPUT);
  pinMode(EW_YELLOW_PIN, OUTPUT);
  pinMode(EW_GREEN_PIN, OUTPUT);
  Serial.begin(9600);
  enterState(currentState);
}
```

```
void loop() {
  switch (currentState) {
    case STATE_NS_GREEN:
      delay(10000);  // 10 seconds
      currentState = STATE_NS_YELLOW;
      break;
    case STATE_NS_YELLOW:
      enterState(STATE_NS_YELLOW);
      delay(3000);   // 3 seconds
      currentState = STATE_EW_GREEN;
      break;
    case STATE_EW_GREEN:
      enterState(STATE_EW_GREEN);
      delay(10000);  // 10 seconds
      currentState = STATE_EW_YELLOW;
      break;

    case STATE_EW_YELLOW:
      enterState(STATE_EW_YELLOW);
      delay(3000);   // 3 seconds
      currentState = STATE_NS_GREEN;
      enterState(currentState);
      break;
```

```
  }
}
```



Reflections:

The traffic light controller was implemented using an Arduino UNO and a Finite State Machine (FSM) approach.

Four states were designed: North-South Green, North-South Yellow, East-West Green, and East-West Yellow.

The green lights remain ON for 10 seconds, while the yellow lights remain ON for 3 seconds before transitioning to the next state.

The digitalWrite() function controls the traffic signal LEDs, and the Serial Monitor displays the current traffic state.

The experiment demonstrates FSM-based control, state transitions, and traffic signal sequencing through Arduino programming and Proteus simulation.